

AI ACADEMIC MENTOR

Course	CSE 4104 — Software Development III
Team	CSE4104-7C-T05
Project Title	AI Academic Mentor
Instructor	Md. Riaz Mahmud, Assistant Professor, Dept. of CSE, NUBTK
Section	7C
Submission Date	29 June 2026
GitHub Repository Link	cse4104-7c-t05-ai-academic-mentor/documentation at main · Priyo-1158/cse4104-7c-t05-ai-academic-mentor · GitHub

Team Information

Role	Name	Student ID	Contribution
Team Leader	Shadman Sadequeen Priyo	11230121158	Project coordination, GitHub management, integration testing
Frontend Developer	Musabbir Hossain Chayon	11230121168	API integration, Report, UI pages, Netlify deployment, Next.js
Backend Developer	SK. Nadirul Haque Nadir	11230121155	Node.js, Express, routing, Express routes, MongoDB models, JWT authentication
AI Integration Lead	Shafin Kabir	11230121175	Gemini API, HuggingFace, AI routes

DATABASE DESIGN & IMPLEMENTATION

Database Platform

MongoDB Atlas (cloud-hosted) is used as the primary database. Mongoose 8.0.3 provides schema enforcement, validation, and ODM functionality on top of raw MongoDB. The system also includes a full in-memory fallback mode that activates automatically when MongoDB is unavailable — ensuring the server never crashes during development or demo.

Collections Overview

Collection	Model	Primary Key	Purpose
users	User	_id (ObjectId)	Student & admin accounts
quizresults	QuizResult	_id (ObjectId)	Quiz attempt scores & answers
studypplans	StudyPlan	_id (ObjectId)	AI-generated study schedules
notes	Note	_id (ObjectId)	User notes & AI summaries

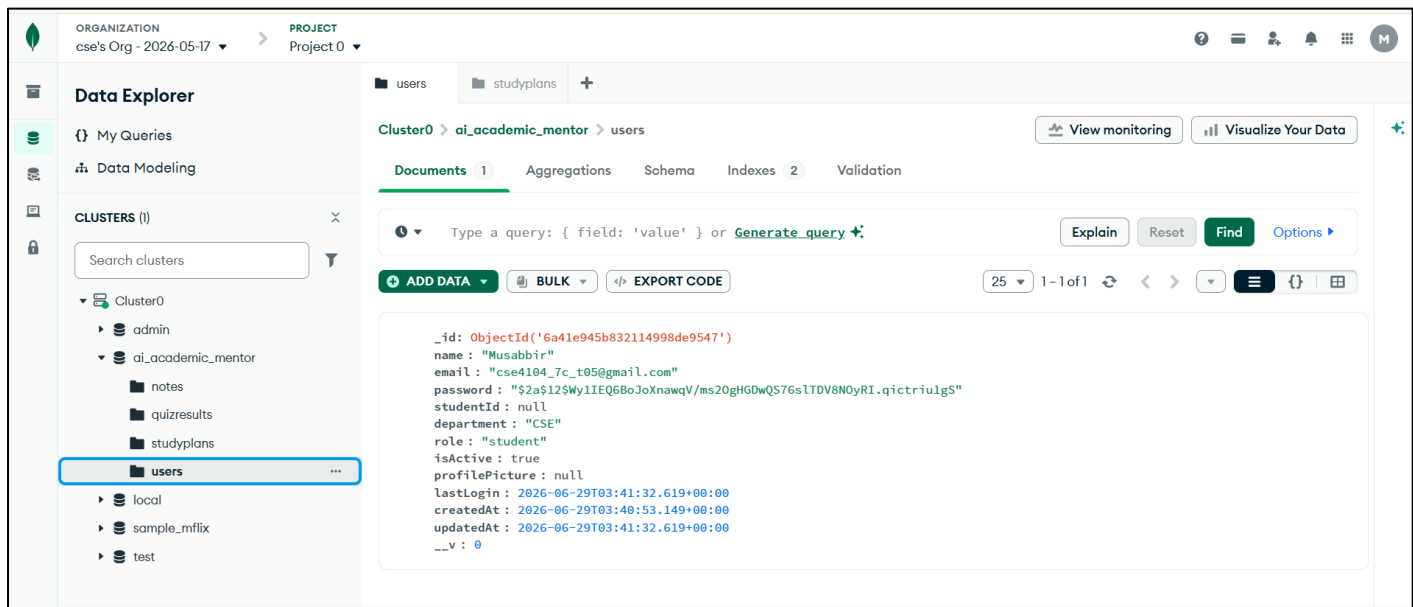


Figure : Collections Overview

Collection Schemas

users Collection

Field	Type	Required	Description / Validation
_id	ObjectId	PK	Auto-generated primary key
name	String	YES	2–100 characters, trimmed
email	String	YES	Unique, lowercase, regex validated
password	String	YES	bcrypt hash (cost 12), select:false
studentId	String	NO	Optional student ID
department	String	NO	Default: 'CSE'
role	Enum	NO	'student' 'admin' (default: student)
isActive	Boolean	NO	Account status (default: true)
lastLogin	Date	NO	Updated on each successful login
createdAt / updatedAt	Date	AUTO	Mongoose timestamps

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - ai_academic_mentor
 - notes
 - quizresults
 - studyplans
 - users**
 - local
 - sample_mflix
 - test

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA **BULK** **EXPORT CODE**

```

_id: ObjectId('6a41e945b832114998de9547')
name: "Musabbir"
email: "cse4104_7c_t05@gmail.com"
password: "$2a$12$Wy1IEQ6BoJoXnawqV/ms20gHGDwQS76slTDV8N0yRI.qictriu1gS"
studentId: null
department: "CSE"
role: "student"
isActive: true
profilePicture: null
lastLogin: 2026-06-29T03:41:32.619+00:00
createdAt: 2026-06-29T03:40:53.149+00:00
updatedAt: 2026-06-29T03:41:32.619+00:00
__v: 0
  
```

Figure : Users Collection

quizresults Collection

Field	Type	Required	Description
_id	ObjectId	PK	Auto-generated
user	ObjectId	FK	Reference → users._id (indexed)
topic	String	YES	Quiz subject/topic name
score	Number	YES	Percentage score (0–100)
numQuestions	Number	YES	Total questions in quiz
correct	Number	YES	Count of correct answers
difficulty	Enum	NO	easy medium hard
type	Enum	NO	mcq truefalse mixed
answers	Mixed	NO	Full answer log array

The screenshot shows the MongoDB Compass interface. On the left, the 'CLUSTERS (1)' sidebar lists a cluster named 'Cluster0' with several databases. The 'ai_academic_mentor' database is expanded, showing collections: 'notes', 'quizresults' (highlighted), 'studyplans', and 'users'. The main panel on the right shows a query result for the 'quizresults' collection. It includes a search bar, buttons for 'ADD DATA', 'BULK', and 'EXPORT CODE', and two document snippets. Each snippet shows fields like _id, user, topic, score, numQuestions, correct, difficulty, type, answers, createdAt, updatedAt, and __v.

```

_id: ObjectId('6a4171e87bca54ef5c0332e4')
user: ObjectId('6a41408d7f28104d8d479bef')
topic: "Database Management System"
score: 80
numQuestions: 5
correct: 0
difficulty: "medium"
type: "mcq"
answers: Array (empty)
createdAt: 2026-06-28T19:11:36.100+00:00
updatedAt: 2026-06-28T19:11:36.100+00:00
__v: 0

_id: ObjectId('6a41edd8b832114998de9552')
user: ObjectId('6a41e945b832114998de9547')
topic: "Data Structures"
score: 80
numQuestions: 5
correct: 4
difficulty: "medium"
type: "mcq"
answers: Array (empty)
createdAt: 2026-06-29T04:00:24.295+00:00
updatedAt: 2026-06-29T04:00:24.295+00:00
__v: 0

```

Figure : QuizResults Collection

studyplans Collection

Field	Type	Required	Description
_id	ObjectId	PK	Auto-generated
user	ObjectId	FK	Reference → users._id (indexed)
title	String	NO	Plan title (default: 'My Study Plan')
subjects	[String]	NO	Array of subject names
days	[Object]	NO	Array of day sub-documents with sessions
goal	String	NO	revision exam practice
hoursPerDay	Number	NO	Daily study hours (default: 3)
level	Enum	NO	beginner intermediate advanced

CLUSTERS (1)

- Cluster0
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Type a query: { field: 'value' } or [Generate query](#)

ADD DATA
BULK
EXPORT CODE

```

_id: ObjectId('6a41ee70b832114998de9554')
user : ObjectId('6a41e945b832114998de9547')
title : "Operating System 3-Week Plan"
subjects : Array (empty)
days : Array (empty)
goal : "revision"
hoursPerDay : 3
level : "intermediate"
isActive : true
createdAt : 2026-06-29T04:02:56.653+00:00
updatedAt : 2026-06-29T04:02:56.653+00:00
__v : 0

```

Figure : StudyPlans Collection

notes Collection

Field	Type	Required	Description
_id	ObjectId	PK	Auto-generated
user	ObjectId	FK	Reference → users._id (indexed)
title	String	YES	Max 200 characters
originalText	String	YES	Raw user note content
summary	String	NO	AI-generated summary text
subject	String	NO	Subject/course tag
style	Enum	NO	concise detailed bullets exam
tags	[String]	NO	Custom tags array

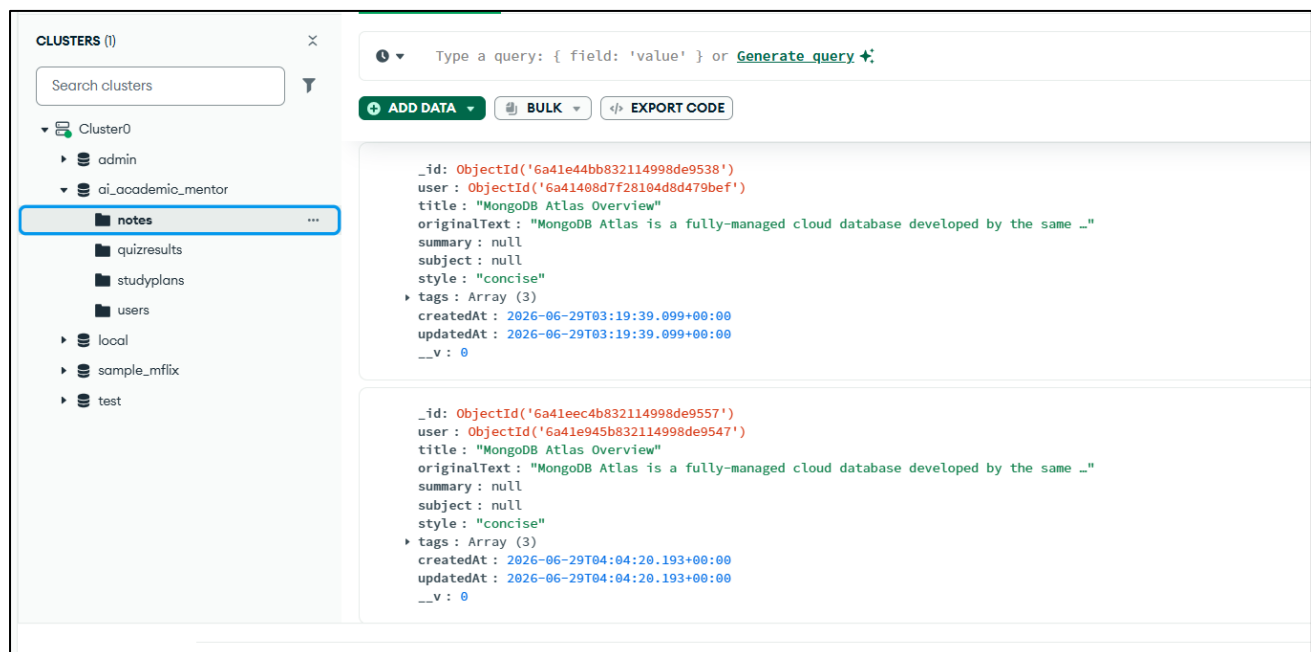


Figure : Notes Collection

Entity Relationships

From	Type	To	FK Field
users (1)	One-to-Many	quizresults	quizresults.user → users._id
users (1)	One-to-Many	studyplans	studyplans.user → users._id
users (1)	One-to-Many	notes	notes.user → users._id

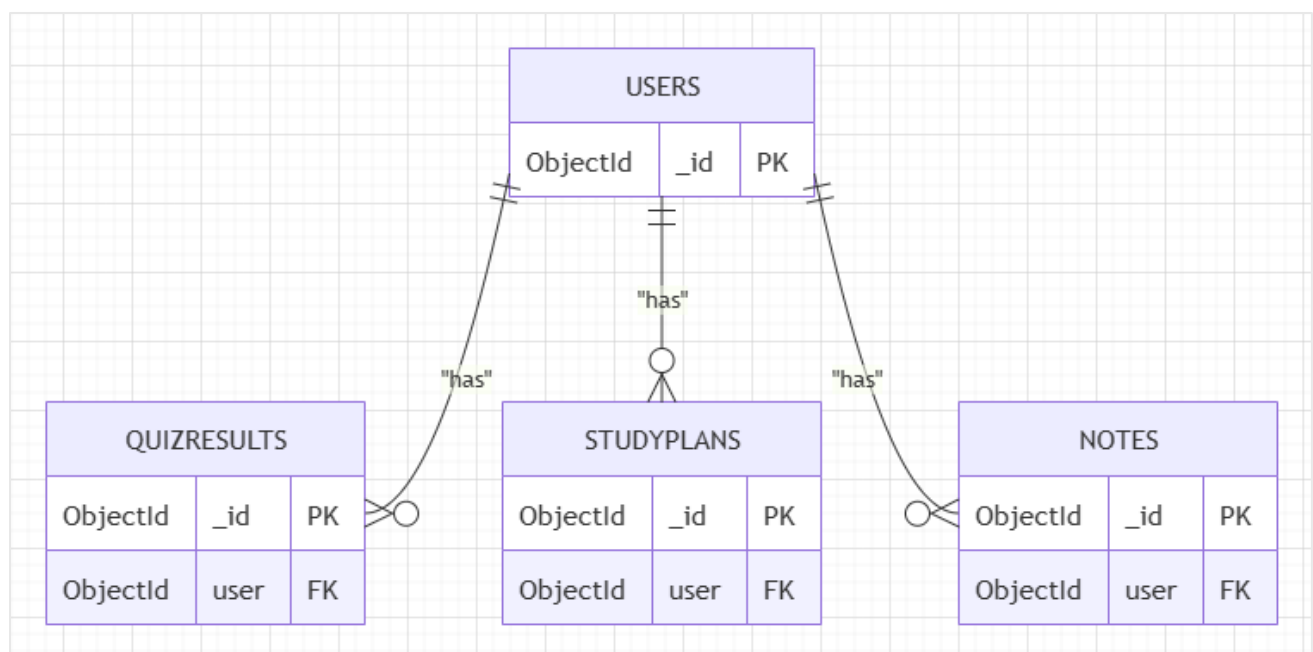


Figure : ER Diagram